

Assistant: Hello! I'm an AI assistant here to help you narrow down the Material requirements for Grade selection. Could you please tell me which application you're considering?

User: I am looking for a material for a CCM - fixed focus

Assistant: To help you select the best material for your Compact Camera Module (CCM) - fixed focus application, could you please clarify:

1. What is the exact use of the polymer in your CCM? For example, is it for the housing, base, holder, cover, or another component?
2. What is the end-use device for this CCM? Is it for a smartphone, tablet, security camera, or another electronic device?

This information will help me understand your requirements and suggest suitable material options.

User: It is a robotic device. It is a sensor connector. Needs to be antistatic.

Assistant: To help identify the best material for your sensor connector in the robotic device, could you please provide:

1. What are the operating conditions for the connector? For example, is it exposed to internal heat, vibration, or occasional maintenance? (Based on typical robotic device environments, I assume moderate heat, vibration, and ESD risk.)
2. What is the minimum wall thickness of your connector part?
3. Have you used any specific base materials for this connector in previous designs, or do you have a preferred polymer?

4. Are you open to alternative material suggestions, or do you require the same base material as before?

This will help ensure the material meets your technical and processing requirements.

User: i dont have this information and no preference

Assistant: Here are the key material requirements (CTQs) assumed for your sensor connector application:

1. Flammability Classification (UL 94): V-0 at 0.8 mm
2. Minimum Thickness: 0.8 mm
3. Relative Thermal Index - Electrical (RTI Elec): ≥ 120 °C
4. Antistatic: Required (surface resistivity in the static-dissipative range 10^6 – 10^9 ohm/sq)
5. Electrical insulation: Required
6. Injection Molding: Required
7. High Flow (for thin-wall molding): Relevant
8. Dimensional stability / Low warpage: Relevant
9. Good weld-line performance: Relevant
10. Vibration resistance / adequate stiffness: Relevant
11. Flame retardant: Required

Are all of these requirements correct for your application? Please confirm if these match your needs, or let me know if any should be changed.

User: dimensional stability is very important. I need built in antistatic performance.

It also needs good adhesion to metal. RTI or UL94.

and recommend the grades based on these

Assistant: Based on your requirements for dimensional stability, built-in antistatic performance, good adhesion to metal, and compliance with RTI and UL94, the following grades may be suitable for your sensor connector application:

1. ****VECTRA® A725****

- Built-in antistatic (static-dissipative) performance: surface resistivity $1 \times 10^6 \Omega$
- UL 94 V-0 at 0.8 mm
- RTI Electrical: 130°C
- High flow, excellent dimensional stability, and good weld-line performance
- Suitable for thin-wall injection molding and provides reliable adhesion to metal with proper surface preparation

2. ****VECTRA® A700****

- Built-in antistatic (static-dissipative) performance: surface resistivity $\sim 1 \times 10^6 \Omega$
- UL 94 V-0 at 0.42 mm (meets/exceeds requirement)
- RTI Electrical: 130°C
- High flow, low warpage, and good stiffness
- Also suitable for thin-wall injection molding and metal adhesion

These grades are potential options based on technical data and your specified requirements. Please ensure compliance with all applicable standards and regulations for your final design. Final material selection and product design decisions should be validated in your application environment.

Note: This recommendation is generated by an AI assistant and should be considered purely informational.